# String Graph Obstacles of High Girth and of Bounded Degree

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# Introduction

- String graphs are the intersection graphs of curves in the plane
- Originally studied by Sinden [4] in relation to RC circuits

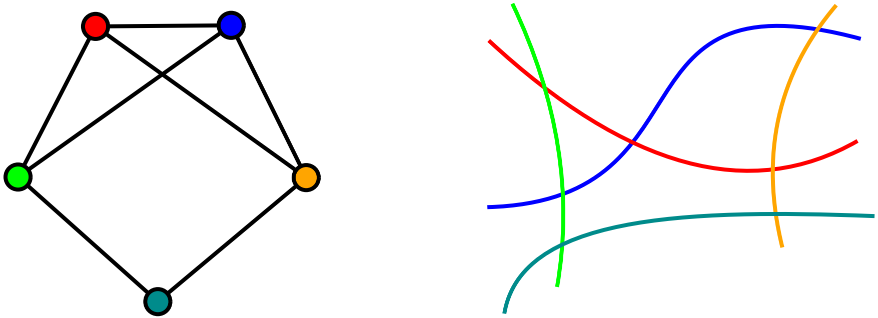
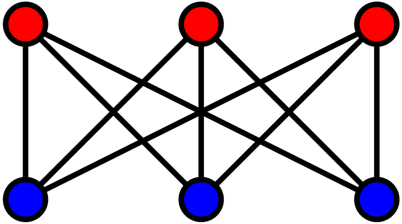


Figure: A graph and a corresponding string representation.

# Relation to planar graphs

- Every planar graph is a string graph
- Not every string graph is a planar graph (for example, $K_t$ and $K_{t,t}$ are string graphs for all $t \in \mathbb{N}$)
- Planar graphs are closed under taking minors
- String graphs are closed under taking *induced* minors



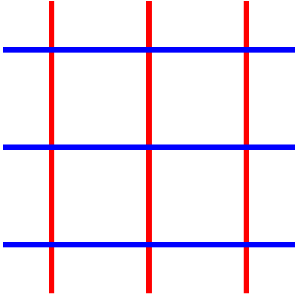


Figure: A non-planar graph $K_{3,3}$ and a corresponding string representation.

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**Theorem (Kratochvíl [2])**

*There are infinitely many string graph obstacles.*

**Theorem (Kratochvíl [3])**

*The string graph recognition problem is NP-hard.*

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- We investigate two cases: the triangle-free case and the maximum degree 3 case

# The triangle-free case

We prove:

**Theorem**

*There are infinitely many string graph obstacles of girth* $4$.

# The necklace graph $N_i$

- For $i \in \mathbb{N}$, the necklace graph $N_i$ consists of:
    - $i$ *links*, each consisting of a four-cycle of two non-adjacent white vertices, one red vertex, and one blue vertex
    - One *pendant*, which is isomorphic to $\text{sub}_1(K_{1,3})$
    - All pendant vertices are white except for the *jewel*, which is both red and blue

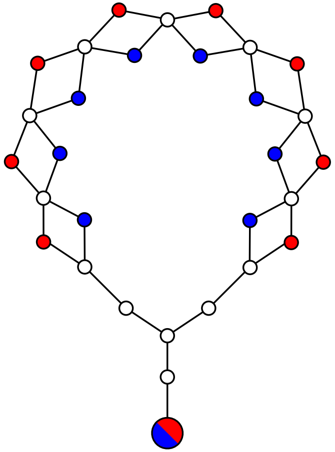
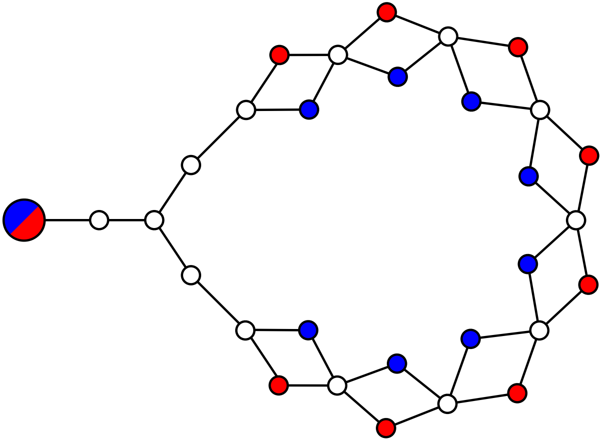
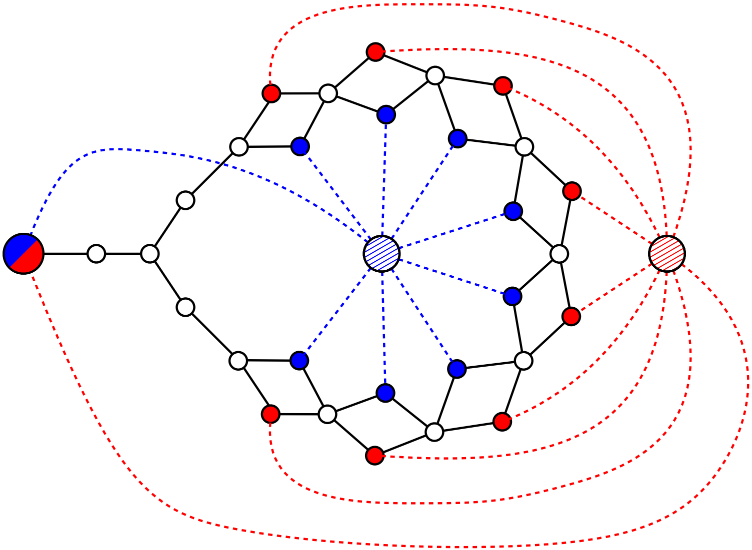


Figure: The necklace graph $N_8$.

# $N_8$



# $\hat{N}_8$



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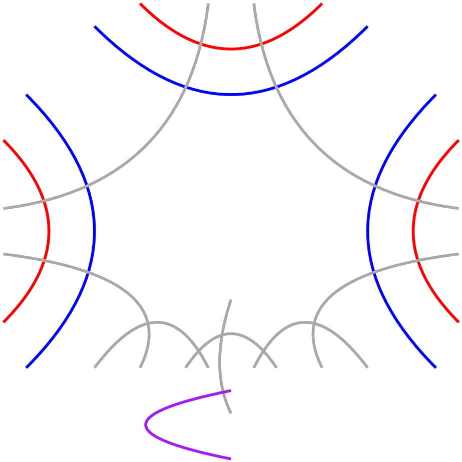
For each $i \geq 3$ we need to prove:

1. $\hat{N}_i$ is not a string graph
2. For any $v \in V(\hat{N}_i)$, $\hat{N}_i \setminus v$ is a string graph
3. For any $e \in E(\hat{N}_i)$, $\hat{N}_i / e$ is a string graph

# Proof idea for 1



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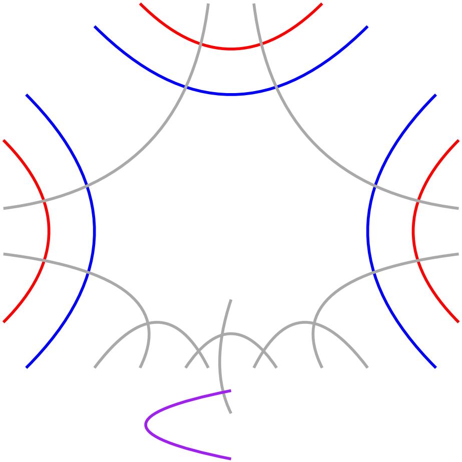

- Each apex string blocked by cycle of opposite color

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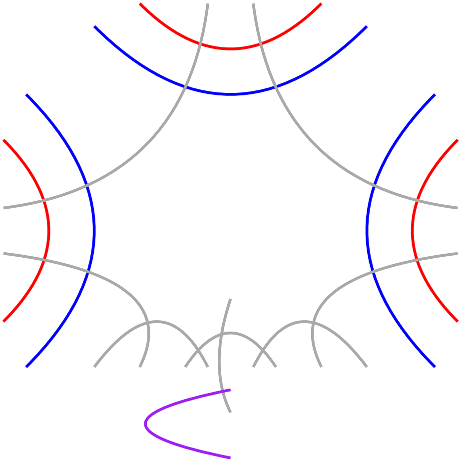

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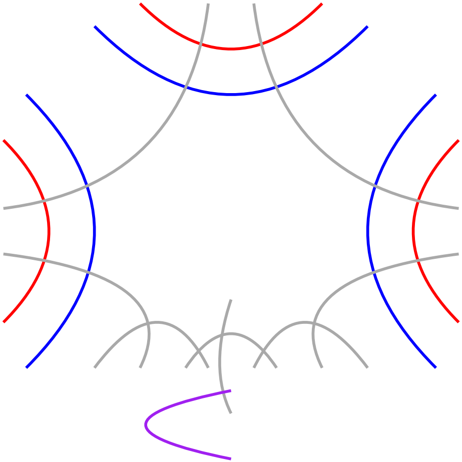

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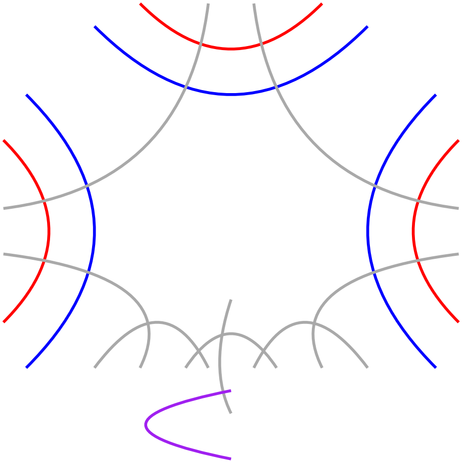
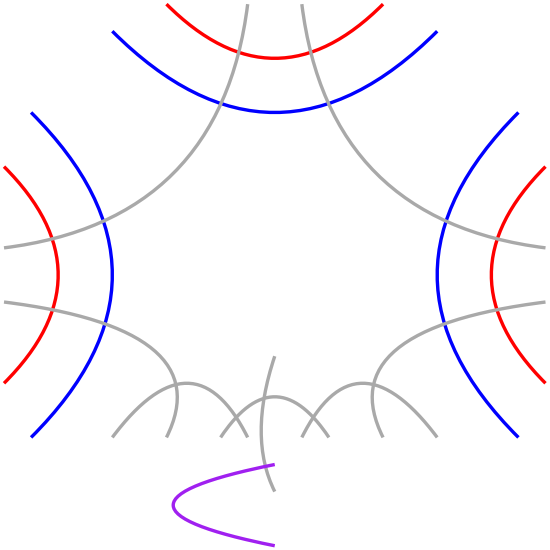


- Each apex string blocked by cycle of opposite color
- Each apex string needs to intersect the jewel string
- Each apex string needs to intersect all other strings of its color
- When $i \geq 3$ the apex strings unavoidably block each other

# Proof idea for 2 and 3 (example: edge contraction)



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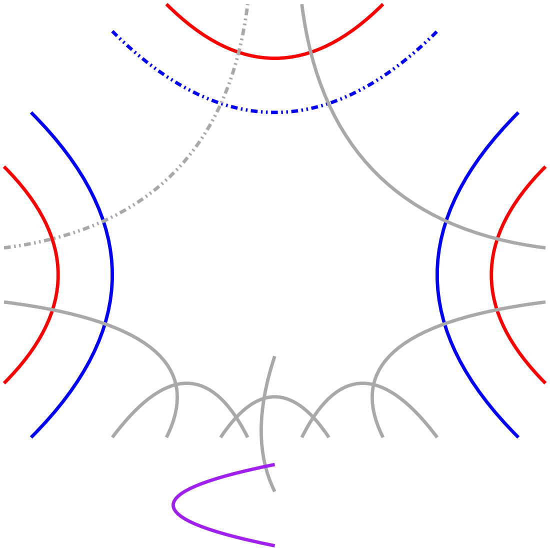

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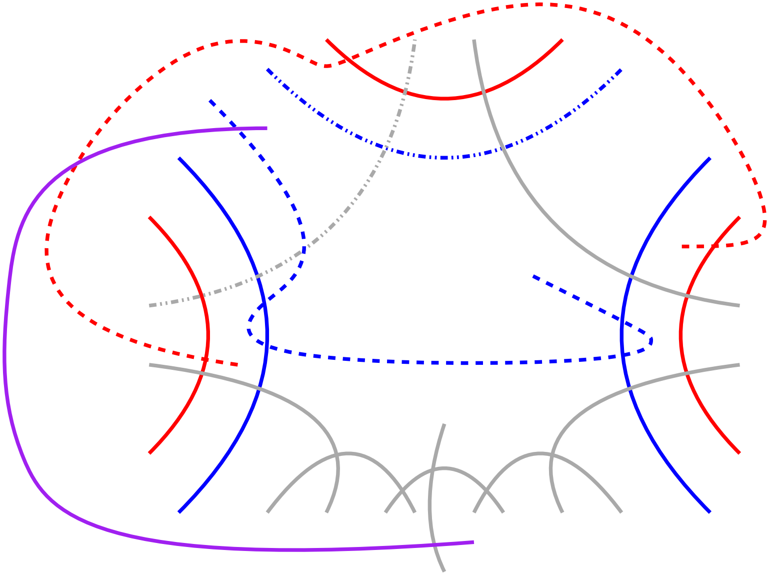

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*A subcubic graph $G$ is a string graph if and only if there exists a cubic matching $M$ in $G$ such that $G/M$ is planar.*

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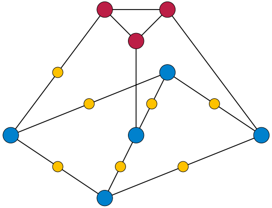

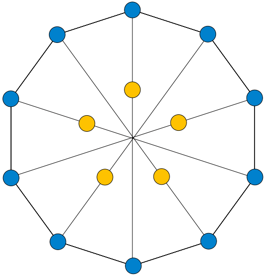

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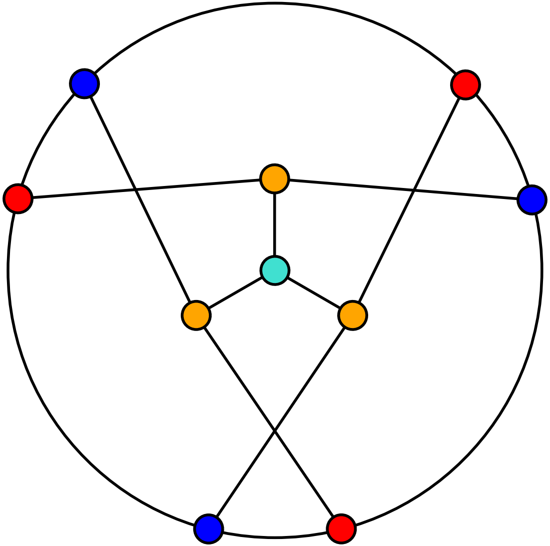

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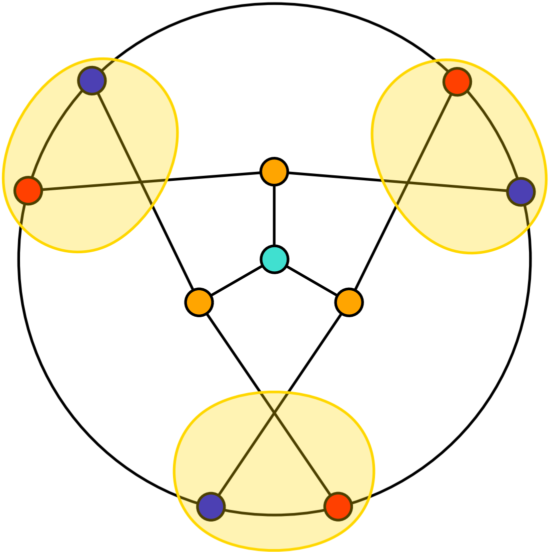

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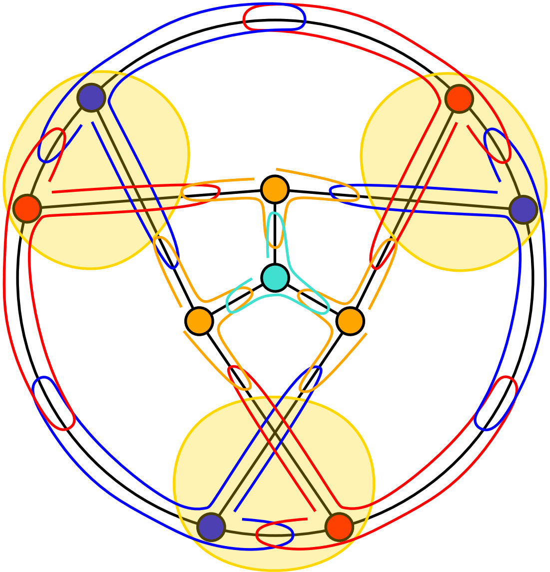

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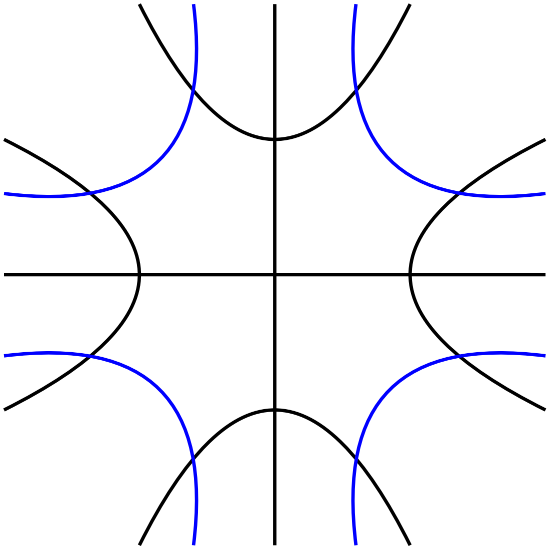

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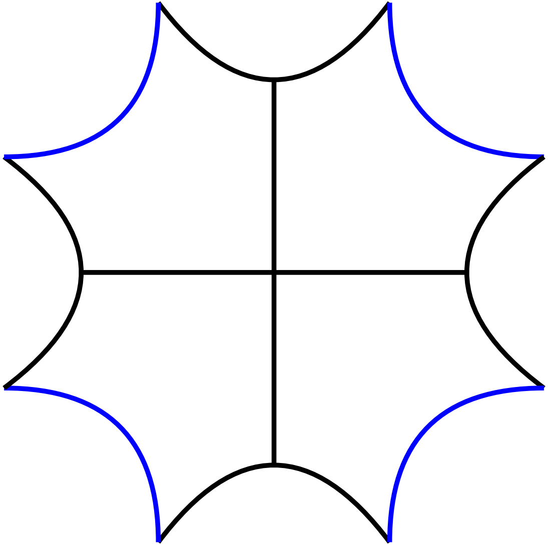

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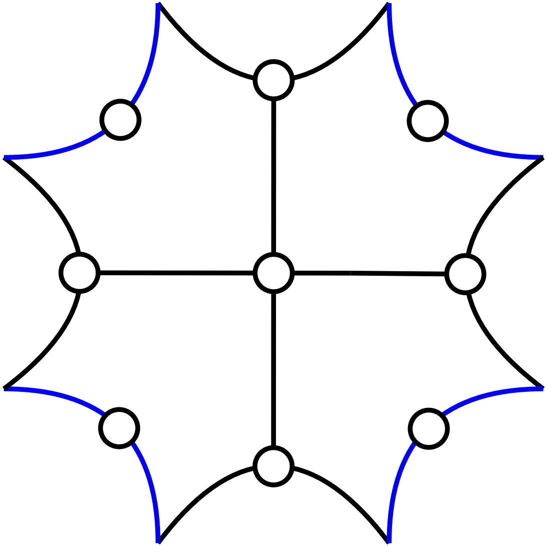

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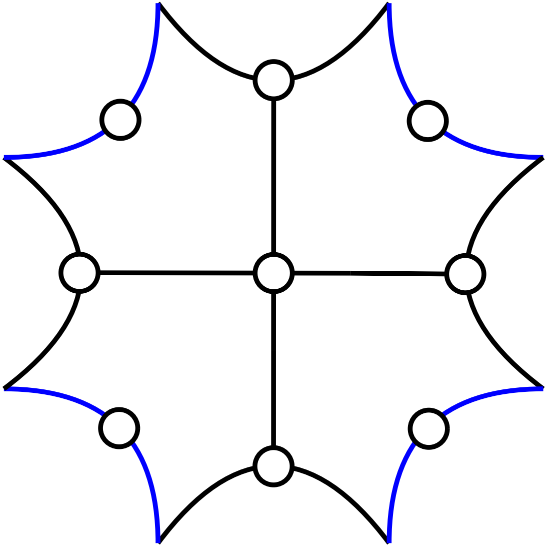

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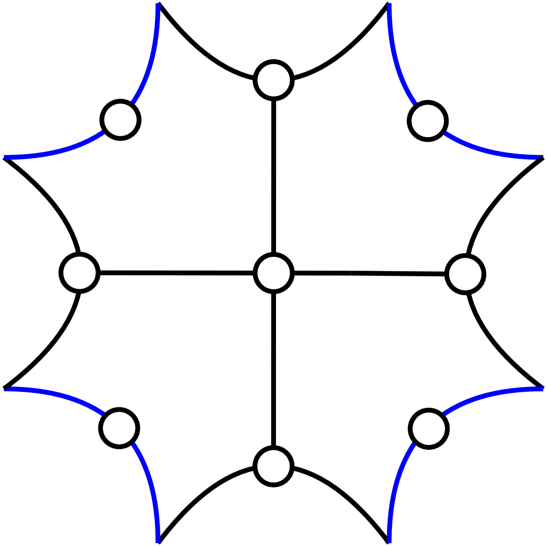

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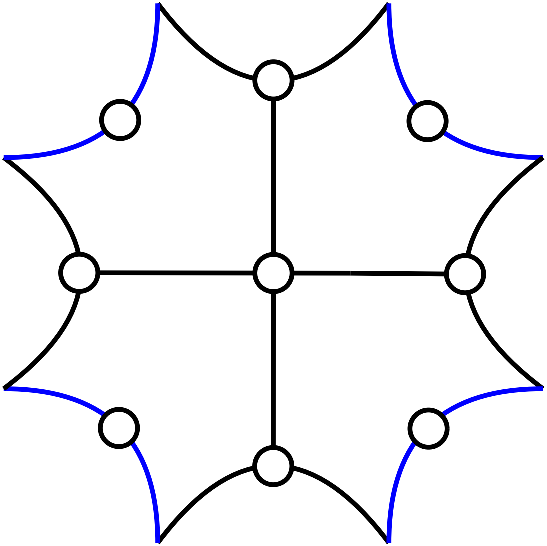

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- For each string $S$ of degree 3, draw a vertex at the point $p$ on $S$ that is not an endpoint of $S$ in which another string intersects $S$
- This *almost* yields a planar drawing of $G$; but strings that properly cross have their vertices drawn in the same place!

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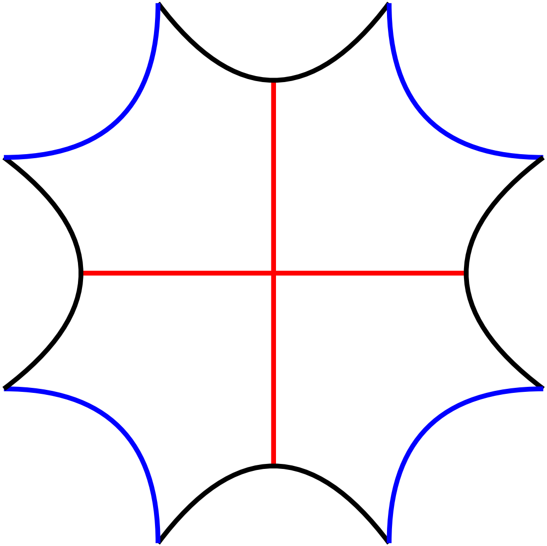

- We have constructed a planar drawing of the graph obtained from $G$ by contracting the edge between each pair of properly crossing strings
- Any such pair consists of two degree 3 vertices
- This set of edges forms a matching of $G$

# Corollaries

## Recognition

The string graph recognition problem on subcubic graphs is fixed-parameter tractable in treewidth.

## Girth of subcubic obstacles

Every subcubic obstacle has girth smaller than 30.

# References I


[1] Gideon Ehrlich, Shimon Even, and Robert E. Tarjan. "Intersection graphs of curves in the plane". In: *J. Combin. Theory* 21.1 (1976), 8–20. DOI: 10.1016/0095-8956(76)90022-8.

[2] Jan Kratochvíl. "String Graphs. I. The number of critical nonstring graphs is infinite". In: *J. Combin. Theory Ser. B* 52.1 (1991), 53–66. DOI: 10.1016/0095-8956(91)90090-7.

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